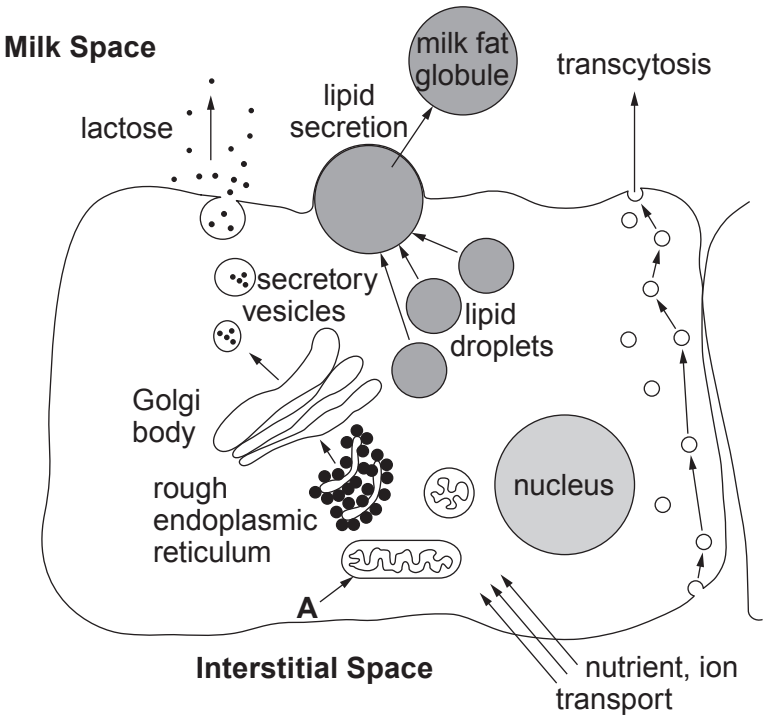


5. Human milk contains all the nutrients a young baby needs in exactly the right proportions. It is formed in the mammary glands by small groups of milk-producing cells in structures called lobules. These cells absorb molecules from the blood and use them to synthesise the lipids, carbohydrates and proteins found in milk. **Image 5.1** shows a milk producing cell.

Image 5.1



Lactose is synthesised in the Golgi body and is transported in vesicles through the cytoplasm. The vesicles increase in diameter as they move towards the plasma membrane. The vesicle membranes fuse with the plasma membrane and the vesicles empty their contents out of the cell. Lipids are also secreted in droplets.

(a) Use the information provided and your own knowledge to answer the following questions.

- (i) State the name of the process by which the vesicles empty their contents out of the milk-producing cell. [1]

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- (ii) Identify the organelle labelled **A** and explain its role in the function of this cell. [2]

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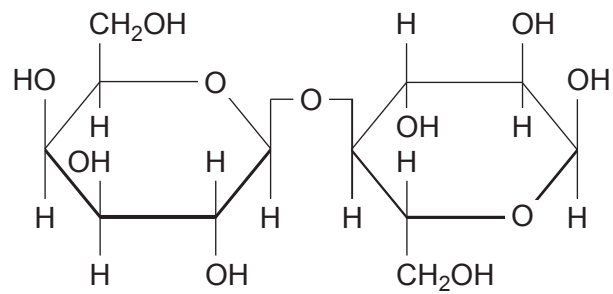
- (iii) Use **Image 5.1** to suggest how the cell avoids a continual increase in the area of its cell surface membrane. [1]

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(b) The main carbohydrate in milk is lactose. **Image 5.2** shows a molecule of lactose.

Image 5.2



(i) **Complete Image 5.2** to show the digestion of lactose. [2]

(ii) State:

I. the type of reaction; [1]

.....

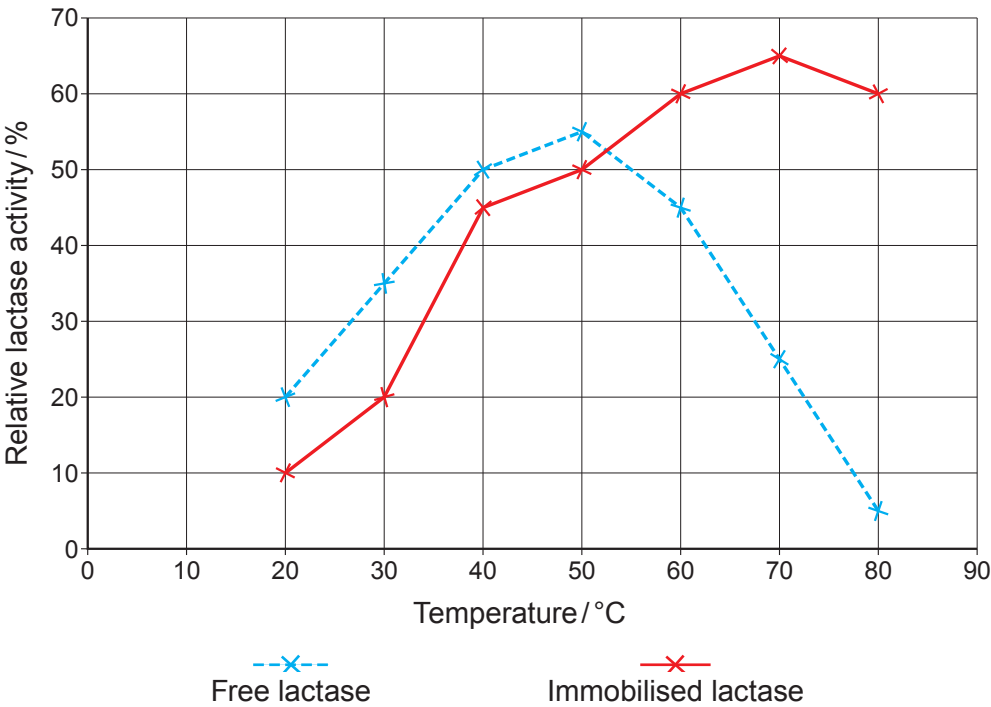
II. the names of the products of digestion of lactose. [1]

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- (c) Some people are lactose intolerant and in recent years it has been possible to produce lactose free milk and milk products using the enzyme lactase. One way to do this is to pour the milk through a column of alginate beads containing immobilised lactase. **Graph 5.3** shows the activity of free lactase and immobilised lactase at different temperatures.

Graph 5.3



- (i) Using **Graph 5.3** and your own knowledge, compare the relative lactase activity of the free and immobilised lactase at the following temperatures. Explain each answer.

I. Below 50 °C.

[2]

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II. Above 50 °C.

[3]

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(ii) Lactose is a reducing sugar. Explain why it is not possible to use Benedict's reagent to monitor the progress of the digestion of lactose by lactase.

[1]

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